Zero waste concept

Transition to zero waste lifestyle

- **Zero Waste**

- Maximizes recycling
- Minimizes waste
- Reduces consumption
- Ensures that products are made to be reused, repaired or recycled back into nature or the marketplace

- Zero waste is **aimed at eliminating all discharges to land, water or air** that are a threat to planetary, human, animal or plant health

Contd...

- Zero waste is a concept for a range of measures aimed at mimicking (copying) nature's successful strategies to make our industrial systems work
- But moving towards zero waste is a big challenge to humanity.
- For a successful zero waste initiative, waste should be considered as a resource.

Benefits of Zero wastes

Economic benefits

- Reduction of waste can in turn **reduce handling, transport and treatment costs** incurred for solid wastes

Faster Progress

- A zero-waste strategy improves the **production processes and improving environmental prevention** strategies which can lead to take larger, more innovative steps

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Supports sustainability

- A zero-waste strategy supports all three of the generally accepted goals of sustainability- economic well-being environmental protection, and social well-being

Improved material flows

- A zero-waste strategy would use far fewer new raw materials and send no waste materials to landfills
- Any material waste would either return as reusable or recycled materials or would be suitable for use as compost

The conservation of all resources by means of responsible production, consumption, reuse, and recovery of products, packaging, and materials without burning and with no discharges to land, water, or air that threaten the environment or human health.

The [redacted]e is
to push economies towards
the target of sending no
waste to landfill,
incinerators, and the ocean.

Zero waste refers not only
to keeping waste out of
landfill, but also pushing
our economy to be less
wasteful in production and
[redacted]

The goal of Zero Waste is to:

Maximize recycling
Minimize waste
Reduce consumption
Ensure products are made to be reused, repaired, or recycled
Purchase sustainable products

How it helps?

Saving energy - especially by reducing energy consumption associated with extracting, processing, and transporting raw materials and waste

Reducing and eventually eliminating the need for landfills and incinerators

Underlying obligations of the three target sections of society

Producers are at the front end, and they must take responsibility for product design and

Political responsibility must bridge the gap between community and producer, promoting both environmental and human health while enforcing new laws

The Community sits at the back end, taking responsibility for consumption and disposal

Zero Waste Principles

- Design closed-loop systems
- Ensure processes (manufacturing, recycling, etc.) happen close to the source
- Conserve energy
- Don't export harmful waste
- Engage the community and promote change
- Keep products and materials in the loop as long as possible
- Develop adaptable, flexible, and resilient systems

- Build systems that provide feedback for continuous improvement
- Support local economies
- Promote materials as resources
- Minimize polluting discharges to land, water, and air
- Consider the true costs of opportunities
- Promote the Precautionary Principle
- Promote the Polluter Pays Principle

Zero waste

How San Francisco Is Becoming A Zero Waste City - YouTube

https://www.youtube.com/watch?v=8nr1gzGdfx0

Circular Economy

- An economic model that aims to reduce waste and reuse resources
- In contrast to the 'produce-consume-waste', linear model, a circular economy is based on a restorative or regenerative design of industries, businesses, processes etc.
- This minimizes/eliminates the waste and promotes sustainability throughout the life cycle
- Treating waste as a potential resource and recovering materials from waste streams through recycling might generate revenue and make the value chain self-sustaining

Circular economy

- Natural resource extraction and processing account for approximately half of global greenhouse gas emissions
- A circular economy approach reduces materials use, redesigns materials to be less resource intensive, and recaptures “waste” as a resource that can serve as feedstock to manufacture new materials and products
- Circular economy refers to an economy that uses a systems-focused approach and involves industrial processes and economic activities that are restorative or regenerative by design, enable resources used in such processes and activities to maintain their highest value for as long as possible, and aim for the elimination of waste through the superior design of materials, products, and systems
- Essential to reducing greenhouse gas

HAZARDOUS WASTE

- Hazardous wastes are those that can cause harm to human and the environment.

Characteristics of Hazardous Wastes:

- **Toxic wastes:**

Toxic wastes are those that are poisonous in small or trace amounts.

- **Reactive wastes:**

Reactive wastes are those that have a tendency to react vigorously with air or water are unstable to shock or heat, generate toxic gases or explode during routine management.

- **Infectious wastes:**

Included human tissue from surgery, used bandages and hypoderm needles hospital wastes.

- **Corrosive wastes:**

Are those that destroy materials and living tissues by chemical reactions

Hazardous and e-waste identification

1. Listed Waste by US Environmental Protection Agency (US-EPA)

a) F-List

b) K-List

c) P-List

d) U-List

2. Characterized Waste

a) Ignitability

b) Corrosivity

c) Reactivity

d) Toxicity

Contd....

3. Universal Waste

a) Batteries

b) Pesticides

c) Equipment containing mercury

d) Lamps containing mercury

4. Mixed Waste: Radioactive + Hazardous

5. E-Waste: Electrical Waste, Electronic Waste

Hazardous and e-waste identification

- Electronics waste, commonly known as e-scrap or e waste, is the trash we generate from surplus, broken, and obsolete electronic devices
- Electronics contains various toxic and hazardous chemicals and materials that are released into the environment if we do not dispose of them properly
- E-waste or electronics recycling is the process of recovering material from old devices to use in new products

E~waste (Electronic~waste) management

E~waste Management

- E~waste management should begin at the point of generation
- This can be done by:~

Waste minimization techniques by sustainable product design

Steps for E-waste management

1. Use 3R technique to minimize the E-waste
2. Identify the e-waste category item
3. Identify the e-waste composition or determine it
4. Identify possible hazardous content in e- waste
5. Identify, whether the e-waste component is hazardous or the entire E-waste item is hazardous

Hazardous waste management

- Hazardous waste can be treated by chemical, thermal, biological, and physical methods
- Chemical methods include ion exchange, precipitation, oxidation and reduction, and neutralization
- Among thermal methods is high-temperature incineration, which not only can detoxify certain organic wastes but also can destroy them
- Special types of thermal equipment are used for burning waste in either solid, liquid, or sludge form

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- Hazardous wastes that are not destroyed by incineration or other chemical processes need to be disposed of properly
- For most such wastes, land disposal is the ultimate destination, although it is not an attractive practice, because of the Environment risk is involved
- A common type of temporary storage impoundment for hazardous liquid waste is an open pit or holding pond, called a lagoon
- New lagoons must be lined with impervious clay soils and flexible membrane liners in order to protect groundwater
- Leachate collection systems must be installed between the liners, and groundwater monitoring wells are required